

DATA CLEANING & VISUALIZATION

TRANSFORMING RAW DATASETS INTO
USEFUL INFORMATION





PHASE I:

THE CLEANING PIPELINE

**TURNING 'MESSY' DATA INTO RELIABLE
SOURCE FOR ANALYSIS**



INITIAL DATA ASSESSMENT:

THE CHALLENGE:

THE ORIGINAL MESSY_DATASET.CSV CONTAINED NON-NUMERIC TEXT ENTRIES (E.G., 'THIRTY-EIGHT'), MISSING SALARY VALUES AND INCONSISTENT DATE FORMATS.

TECHNICAL RISKS:

Untreated missing values and duplicates lead to skewed statistics and inaccurate correlation coefficients, undermining decision-making.

The **messy_dataset.csv** file is a small dataset consisting of **10 rows** and **6 columns**.



Core Cleaning Methodology



Standardization

Converting alphanumeric strings to numeric types for Age and Salary.



Imputation

Filling NaNs using the **median** to maintain central tendency.



Deduplication

Removing redundant rows based on core feature combinations.

Age Standardization:

Replaced textual representations with numeric values and converted the entire column to numeric format.

Imputation: Missing values for 'Age' and 'Salary' were filled using their respective **median** values to ensure a robust representation.

ID Maintenance: Missing IDs were filled based on the maximum current ID value to maintain unique identifiers.

De-duplication: duplicate records were removed, ensuring each observation was unique.



Standardizing Age & Salary

Specific operations performed on the dataset:

</> Replace: Mapping text like "thirty-eight" to 38.

📊 Coerce: Converting columns to numeric, shifting non-numerics to NaNs.

🗑️ Median Fill: Ensuring 'Age' and 'Salary' have 100% completion rate.

```
# A. Handle Age: Convert text 'thirty-eight' to 38, handle non-numerics, fill NaNs with median
df_cleaned['Age'] = df_cleaned['Age'].replace('thirty-eight', 38)
df_cleaned['Age'] = pd.to_numeric(df_cleaned['Age'], errors='coerce')
df_cleaned['Age'] = df_cleaned['Age'].fillna(df_cleaned['Age'].median())

# B. Handle Salary: Convert to numeric, fill NaNs with median
df_cleaned['Salary'] = pd.to_numeric(df_cleaned['Salary'], errors='coerce')
df_cleaned['Salary'] = df_cleaned['Salary'].fillna(df_cleaned['Salary'].median())

# C. Handle ID: Ensure numeric and fill missing IDs (max + 1)
df_cleaned['ID'] = pd.to_numeric(df_cleaned['ID'], errors='coerce')
df_cleaned['ID'] = df_cleaned['ID'].fillna(df_cleaned['ID'].max() + 1)

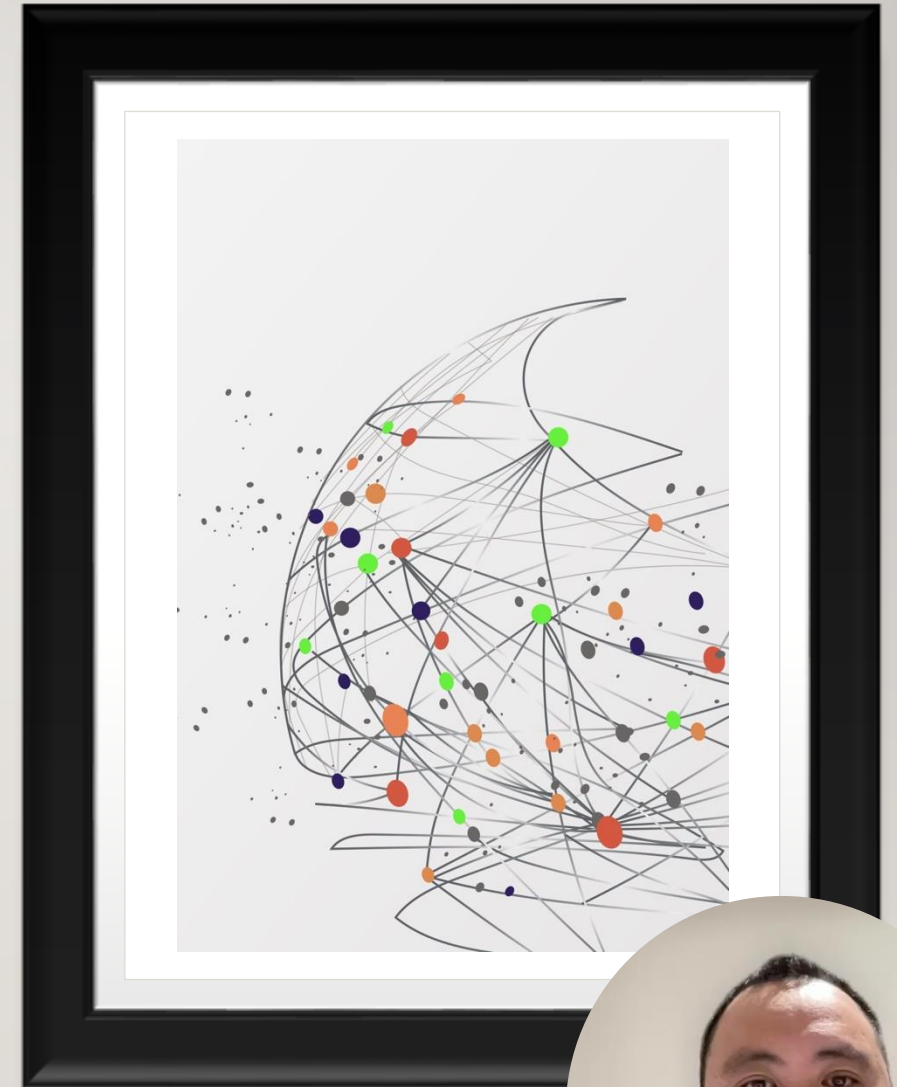
# D. Remove Duplicates: Based on core descriptive columns
df_cleaned = df_cleaned.drop_duplicates(subset=['Name', 'Age', 'Country'])
```



PHASE II:

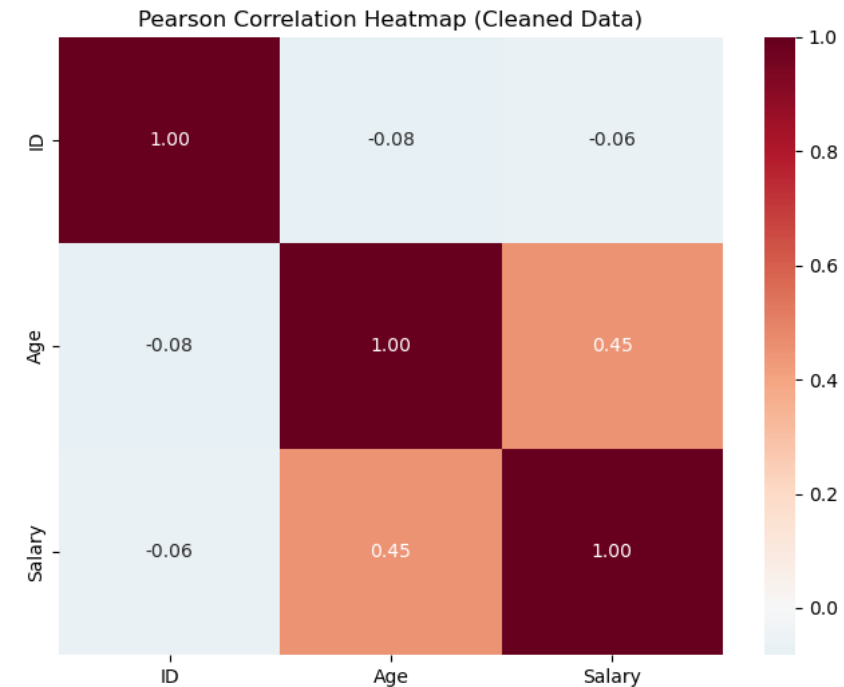
BIVARIATE ANALYSIS

EXPLORING THE LINEAR RELATIONSHIP BETWEEN
AGE AND EARNINGS.

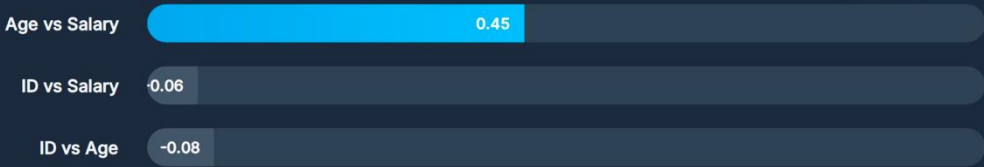


PEARSON CORRELATION ANALYSIS

- The heatmap above shows the linear correlation between numeric features.
- A moderate positive correlation of **0.45** was found between **Age** and **Salary**, suggesting a general trend of higher salaries as age increases within this sample.
- ID** showed almost no correlation with other variables, as expected for an identifier.



Linear Correlation Strengths



Result: A moderate positive correlation (0.45) exists between Age and Salary, confirming that tenure/age generally drives higher income levels in this sample.

Descriptive Statistics

Metric	ID (Index)	Age (Years)	Salary (NZD)
Count	10.0	10.0	10.0
Mean	5.9	30.5	61,600
Standard Deviation	3.6	5.6	5,000
Median (50%)	6.0	29.5	60,000



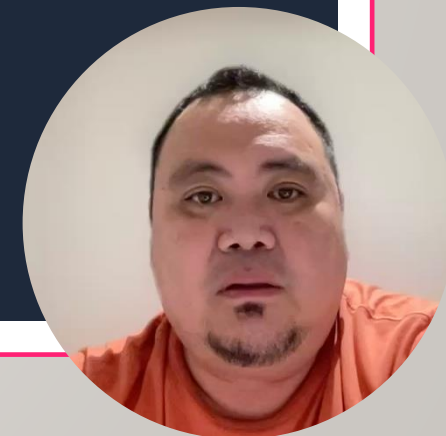
| Outlier Detection

1

Salary
Outlier
Found

The IQR Method

Using the Interquartile Range algorithm ($Q3 - Q1$), we scanned for values exceeding 1.5x the IQR. While Age and ID were clean, the **Salary column** flagged one entry at \$72,000 as a statistical anomaly.



CONCLUSION:

The project successfully transformed a highly unreliable raw dataset into a clean, structured foundation for decision-making. Through a rigorous multi-step pipeline, we achieved several key outcomes:

Integrity Restored: By resolving "messy" elements—such as textual age entries ("thirty-eight"), duplicate "Bob" records, and missing values—we eliminated the risk of skewed statistics that could mislead organizational planning.

Validated Trends: The application of the **Pearson Correlation algorithm** confirmed a **moderate positive relationship (0.45)** between an employee's age and their salary. This validates that tenure and experience generally drive higher income levels within this specific group.

Contextual Outliers: While the **Interquartile Range (IQR)** method flagged one high-earning individual at **\$72,000**, this value was identified as a realistic representation of senior-level pay rather than a data entry error, providing a complete picture of the salary range.

Decision Readiness: The final cleaned dataset—now standardized and imputed using median values to maintain central tendency—provides a 100% completion rate for core features.

Final Takeaway: Strategic data cleaning is not just about "fixing" files; it is a critical prerequisite for accurate bivariate analysis. Without this process, the correlation between age and earnings would have been mathematically impossible to calculate or inherently biased.

